

MuST-VEP: multi-state spliceosome structure for variant effect prediction in spliceosomal snRNA genes

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Every number traces to a committed script; the script number is given in brackets. Primary label set v5c.

Pre-registration: ANALYSIS_PLAN.md, commit 350165e, timestamped before any evaluation reported here.

Abstract

Variants in spliceosomal small nuclear RNA (snRNA) genes have become a frequent identified cause of neurodevelopmental disorder, yet the 2026 classification guidance states that existing in silico tools were not designed or calibrated for non-coding RNA genes and should not be used for PP3/BP4 until they are. The same guidance notes that snRNA structural relationships change through the splicing cycle in ways that may not be captured by analysing a single structure. We address that observation directly. MuST-VEP represents each snRNA nucleotide by its protein and RNA contacts across eleven cryo-EM structures spanning the splicing cycle, projects paralogues and minor-spliceosome analogues onto shared family coordinates, and propagates contact information over a multi-state three-dimensional contact graph; evidence is combined per disease mechanism with the scorer selected by nested leave-one-gene-out cross-validation. Against thirteen baselines under an identical held-out-gene protocol, MuST-VEP reached a mean AUROC of 0.810 for dominant variants, exceeding CADD (0.655), phyloP447 (0.593) and the RNA foundation model RNA-FM (0.460, below chance). Combining all eleven structures outperformed any single splicing state chosen without access to the test gene (0.815 versus 0.631) and exceeded all 200 degree-preserving shuffled-contact nulls. A pre-specified gate failed: gnomAD population constraint was not separable from the model on curated clinical labels (0.766 versus 0.810, CI of the difference -0.123 to $+0.112$). Against measured function, however, the two separated decisively. On 485 RNU4-2 saturation-genome-editing variants scored zero-shot with the entire U4 family withheld, MuST-VEP reached Spearman 0.405 against 0.206 for constraint, 0.072 for CADD and -0.004 for phyloP447; partial correlations were asymmetric, structure retaining 0.394 controlling for constraint while constraint retained 0.025 controlling for structure. Curated pathogenicity labels and molecular function are therefore not interchangeable benchmarks for non-coding variants. Two obvious extensions — conformational-trajectory features and protein-partner identity — were specified with kill criteria, tested, and rejected; both are reported. We provide scores for 13,280 variants across 17 genes, annotated with which loci population sequencing can and cannot assess.

Keywords: spliceosome, small nuclear RNA, variant effect prediction, cryo-EM, neurodevelopmental disorder, saturation genome editing, ACMG calibration

Introduction

Variants in spliceosomal small nuclear RNA genes have moved, in under three years, from a curiosity to one of the more frequent identified causes of neurodevelopmental disorder. *RNU4-2* accounts for roughly 0.4–0.5% of undiagnosed developmental disorder in large cohorts; *RNU2-2* has been described as the most prevalent known recessive neurodevelopmental disorder; *RNU5B-1*, *RNU4ATAC*, *RNU12* and *RNU6ATAC* follow. Together these genes now explain more than one percent of cases that genome sequencing had left unsolved.

Interpreting new variants in them is unresolved. The 2026 guidance for classifying snRNA variants states that existing in silico tools were not designed or calibrated for non-coding RNA genes and should not be used for PP3/BP4 until they are. It also notes that the structural relationships within and between snRNAs change through the splicing cycle, imposing constraints that "may not be captured by the analysis of a single structure".

That second observation is the starting point here. An snRNA is not a static molecule with a fixed set of contacts. U4 is base-paired to U6 in the tri-snRNP, unwound during activation, and absent from the

catalytic spliceosome entirely; U5 loop I holds the exons through both transesterifications; U2 recognises the branch point early. Which nucleotides matter depends on when you look. The spliceosome is, moreover, the one large ribonucleoprotein machine whose entire catalytic cycle has been resolved at near-atomic resolution, so this is a question that can actually be asked. We therefore represent each nucleotide by its contacts across eleven cryo-EM structures spanning the cycle, and ask whether that representation identifies pathogenic variants, predicts measured molecular function, and distinguishes between the alternative alleles at a position.

A second question runs alongside, and it turned out to matter more. Evidence for non-coding variant pathogenicity rests heavily on one source: population sequencing. Curated labels are assembled partly from depletion; the comparator scores are themselves built on conservation and allele frequency. Whether those two things — selection against a variant, and what the variant does to the molecule — are interchangeable has not been tested where both can be measured. snRNA genes are one of the few places where they can, because *RNU4-2* has a saturation genome editing screen that assays function without consulting allele frequency at all.

Three design commitments distinguish this work from the comparable methods literature. First, features, labels, model specification and two gate criteria were frozen in a timestamped analysis plan committed before any evaluation was run; results that failed their criteria are reported as failures, including the one that failed. Second, evaluation holds out an entire gene — strictly harder than the cluster splits usual in this literature, because no variant, position or paralogue of the test gene is seen in training. Third, every claim about structure is tested against a degree-preserving null in which contact profiles are permuted but every node keeps its degree, so a result cannot be produced by the amount of contact alone.

Figures. Figure 1 — the MuST-VEP workflow, the state $\times$ family coverage matrix, and gate G1 against the degree-preserving null (figures/fig1_workflow.png, scripts/54). Figure 2 — the dissociation between clinical labels and measured function (figures/fig2_dissociation.png, scripts/55). Supplementary figures: held-out AUROC per gene, the *RNU4-2* contact profile, and the per-protein mechanism analysis.

Materials and methods

Pre-registration

Features, label set, model specification and the two gate criteria were fixed in a timestamped analysis plan (ANALYSIS_PLAN.md) committed before any of the evaluations reported here were run. Analyses added afterwards are labelled exploratory in the text and in the commit history. No hyperparameter, feature or label was changed after that commit.

Genes, coordinates and paralogue projection

Seventeen spliceosomal snRNA genes were taken from GENCODE v27 [01]. Paralogues and minor-spliceosome analogues were projected onto a shared family coordinate by global pairwise alignment to a family reference ($U4 \rightarrow RNU4-2$, $U6 \rightarrow RNU6-1$, $U5 \rightarrow RNU5A-1$, $U2 \rightarrow RNU2-2P$, $U1 \rightarrow RNU1-1$), so that evidence can transfer between copies (snrna_vep/align.py). Reference sequences are stored with 2,000 bp flanks and are reverse-complemented for minus-strand genes before use.

Structural features

Eleven cryo-EM structures spanning the splicing cycle were used: tri-snRNP 3JCR and 6QW6; pre-B 6QX9; B 5O9Z; Bact 6FF7; Bact-mature 5Z56; C* 5XJC; P 6QDV; U2-snRNP 6Y5Q; minor-Bact 7DVQ; minor-preB 8Y6O. snRNA chains are identified by sequence alignment against the family references rather than by hand-entered chain identifiers, so partially modelled chains still match ($\geq 85\%$ identity over modelled residues). For each nucleotide in each structure we record contacting protein residues, minimum protein distance, snRNA-snRNA contacts and contacts with non-snRNA RNA, using a 4.5 Å heavy-atom cutoff (snrna_vep/contacts.py).

Aggregating over states gives four features: the fraction of resolved states with a protein contact, the maximum number of protein residues contacted, the fraction of states with an snRNA contact, and a graph-smoothed contact fraction obtained by one round of message passing over the multi-state RNA-RNA contact graph (row-normalised adjacency including self-loops and backbone neighbours).

Allele-specific features

Watson-Crick pairing is detected geometrically per structure: a purine N1 within 3.5 Å of a pyrimidine N3 is called a pair and the partner base recorded, so each alternative allele can be classed as preserving a canonical pair (A-U, G-C), forming a wobble (G-U) or breaking the pair. Protein contacts are split into **base** contacts (ring and exocyclic atoms, allele-sensitive) and **backbone** contacts (phosphate and ribose, allele-insensitive), since only base contacts can be perturbed by a substitution (snrna_vep/allele_features.py).

Insertions and deletions are scored at the affected node — the first deleted base, or the anchor+1 base for an insertion — and inherit the position-level features of that node. Allele-specific pairing features are not defined for indels and are not applied to them. This is stated because 70-77% of ReNU cases are a single-base insertion, so the limitation is material.

Labels

Pathogenic variants were curated from nine supplements plus ClinVar (≥ 1 star, 2026-10-04): 266 variants across 10 genes from 14 sources, every row traceable to its source. Controls are mechanism-aware. For recessive genes only gnomAD homozygotes ($nhomalt \geq 1$) and ClinVar B/LB qualify, because heterozygous carriage is uninformative about recessive benignity; *RNU4-2* uses UK Biobank and All of Us variants, which are independent of gnomAD; dominant-only genes use gnomAD PASS alleles restricted to the transcribed region. Homozygous variants from approximately 490,000 UK Biobank genomes were added as recessive controls [05h]. The primary label set is **v5c** — v5 with the 16 variants annotated as recessive-pathogenic yet carried homozygously by a healthy adult removed; v5 is retained as a sensitivity analysis.

Model and evaluation

Features enter as unsupervised within-gene percentile ranks computed over all possible variants of each gene, so no labels are used in feature construction. The model is L2 logistic regression ($C = 0.3$, class-weighted) on [features, features $\times$ recessive-context, recessive-context]. Evaluation holds out an entire gene (*RNU6-1/2/8/9* treated as one unit); a unit and context is evaluated only with ≥ 3 pathogenic and ≥ 3 controls. For each held-out unit the scorer is chosen from four candidates by an inner leave-one-unit-out loop over the **training** units only, so selection never sees the test gene.

Baselines

Thirteen competitors were scored under the identical protocol [23, 53]: gnomAD constraint (1 – observed/expected in 10-nt windows), gnomAD allele density at the position and within ± 2 nt, a trained secondary-structure model (same design and regularisation, contact features replaced by ViennaRNA features), ViennaRNA fold $\Delta\Delta G$, pairing probability and duplex $\Delta\Delta G$, Rfam element-type enrichment, paralogue pathogenic density, distance to the nearest known pathogenic position, CADD (PHRED), phyloP447, and RNA-FM scored zero-shot under the standard language-model convention (damage = $-\log P(\text{alt}) - \log P(\text{ref})$).

Uncertainty

The position-block bootstrap resamples snRNA positions within units, so all variants at a position move together; the hierarchical bootstrap resamples units and then positions. Per-gene AUROCs carry position-block CIs. A logistic GLMM with a gene random intercept is reported as a secondary analysis. Paired comparisons use rank-pooling within unit.

Null model

Per-structure, per-family contact profiles are permuted among resolved positions, preserving each structure's degree sequence; the RNA-RNA contact graph is rewired within each structure so every

node keeps its degree. The full model is refitted on each of 200 replicates.

External data

Saturation genome editing scores for *RNU4-2* from De Jonghe et al. (2026); the *RNU4*ATAC cellular assay and published RNAstructure scores from Benoit-Pilven et al. (2020), whose numbering (NR_023343.1) is 1 nt upstream of the GENCODE model and is shifted accordingly, validated on reference bases at 234/241 SNVs. HPRC Release 2 (v2.1) Minigraph-Cactus pangenome, queried by HTTP range requests against the tabix index (snrna_vep/remote_tabix.py).

Results and discussion

Multi-state structure outperforms any single splicing state

Contact profiles were recomputed from scratch for each structure subset using the main-pipeline definitions, so that configurations differ only in which structures contribute [20, 22]. Combining all 11 cryo-EM structures gave a mean held-out AUROC of 0.815 for dominant variants, against 0.631 for the best single splicing state chosen by nested leave-one-unit-out over the training genes only (position-block Δ AUROC +0.184, 95% CI 0.126–0.292; better in 3 of 4 genes).

No single state suffices because the genes need different ones (Figure 1b). *RNU4-2* requires a pre-activation structure, since every Bact/C/P structure lacks *U4* entirely, whereas **RNU2-2* and *RNU5B-1* are best resolved by activated and catalytic states. Only the multi-state model exceeds 0.73 on all four dominant genes.

Against the degree-preserving null the observed 0.815 exceeded every one of 200 replicates (Figure 1c) (null mean 0.557, 95th percentile 0.689; empirical $p = 0.005$) [22]. All three pre-specified criteria for this gate were met. Removing the minor-spliceosome genes from training left the dominant result unchanged ($0.810 \rightarrow 0.813$), excluding a family-projection artefact [24].

Held-out performance against thirteen baselines

Table 1. MuST-VEP against 13 baselines under the pre-registered held-out-gene protocol.

Baseline	Dominant AUROC	Dominant Δ AUROC [95% CI]	Recessive AUROC	Recessive Δ AUROC [95% CI]
MuST-VEP (this work)	0.810	–	0.662	–
gnomAD constraint (1 – o/e, 10-nt window)	0.766	+0.044 [–0.123, +0.112]	0.658	+0.004 [–0.112, +0.111]
gnomAD allele density (± 2 nt)	0.686	+0.124 [–0.137, +0.182]	0.597	+0.065 [–0.038, +0.172]
Trained secondary-structure model	0.665	+0.145 [+0.068, +0.213]*	0.621	+0.041 [–0.026, +0.116]
CADD (PHRED)	0.655	+0.121 [+0.021, +0.241]*	0.625	+0.037 [–0.083, +0.162]
Paralogue pathogenic density	0.615	+0.195 [+0.136, +0.310]*	0.743	–0.081 [–0.183, +0.013]
gnomAD allele density (position)	0.609	+0.201 [–0.064, +0.259]	0.588	+0.073 [–0.021, +0.169]
phyloP (447-way)	0.593	+0.217 [+0.079, +0.282]*	0.639	+0.023 [–0.035, +0.081]
Distance to nearest known pathogenic position	0.554	+0.256 [+0.185, +0.438]*	0.669	–0.007 [–0.109, +0.086]
Rfam element-type enrichment	0.518	+0.292 [+0.216, +0.363]*	0.618	+0.044 [–0.057, +0.146]
RNA-FM (foundation model, zero-shot)	0.460	+0.350 [+0.143, +0.408]*	0.623	+0.007 [–0.085, +0.095]
ViennaRNA pairing probability	0.434	+0.376 [+0.299, +0.534]*	0.515	+0.147 [+0.017, +0.284]*

Baseline	Dominant AUROC	Dominant Δ AUROC [95% CI]	Recessive AUROC	Recessive Δ AUROC [95% CI]
ViennaRNA fold $\Delta\Delta G$	0.260	+0.549 [+0.353, +0.646]*	0.519	+0.143 [+0.009, +0.277]*
ViennaRNA duplex $\Delta\Delta G$	0.251	+0.598 [+0.194, +0.672]*	0.636	+0.056 [-0.077, +0.191]

Mean AUROC across held-out units. Δ AUROC is MuST-VEP minus the baseline, averaged over units, with a paired position-block bootstrap 95% CI (2,000 resamples). \ marks intervals excluding zero. Hierarchical (unit-then-position) intervals are wider throughout, reflecting that only four units contribute to each context.*

Two results in this table deserve comment. First, **RNA-FM performs below chance on dominant variants** (0.460 at 99.9% coverage). An RNA foundation model trained on sequence cannot rank pathogenic snRNA variants, which is consistent with the guidance's position that general non-coding tools are not calibrated for this gene class, and is worth recording because such models are increasingly offered as general-purpose non-coding predictors.

Second, **gnomAD constraint is not separable from MuST-VEP on clinical labels** (0.766 versus 0.810, CI -0.123 to $+0.112$). This was a pre-specified gate and it failed. We return to it below, because the explanation turned out to be the most consequential result in the study.

In a logistic model with a gene random intercept the odds ratio per standard deviation of score was 3.50 (2.74–4.48), against 1.32 (0.98–1.77) for CADD [24]. For recessive variants MuST-VEP did not beat conservation (Δ AUROC versus phyloP $+0.010$, $p = 0.74$; versus CADD $+0.023$, $p = 0.51$), and no method exceeded 0.69 once healthy homozygous carriers were used as controls. We report this as a limit of position-based prediction rather than a property of any one method: recessive snRNA variants are hypomorphic, cluster in regions that tolerate heterozygous variation, and are measurably milder in the one functional assay available.

Ablation

Table 2. Ablation of every component and of each conformational state.

Ablation	Dominant	Recessive	Δ Dominant
None (full MuST-VEP, 11 structures)	0.806	0.636	-
Remove the contact-graph smoothing term	0.788	0.635	-0.018
Single structure, chosen without the test gene	0.597	0.654	-0.209
Single structure, chosen WITH the test gene (oracle)	0.795	0.675	-0.011
Only 5Z56 (Bact-mature)	0.795	0.644	-0.011
Only 6FF7 (Bact)	0.792	0.634	-0.015
Only 5XJC (C*)	0.752	0.608	-0.054
Only 7DVQ (minor-Bact)	0.747	0.675	-0.060
Only 6QDV (P)	0.687	0.599	-0.120
Only 6QW6 (tri-snRNP)	0.655	0.637	-0.152
Only 6QX9 (pre-B)	0.608	0.604	-0.198
Only 6Y5Q (U2-snRNP)	0.587	0.625	-0.219
Only 8Y6O (minor-preB)	0.582	0.654	-0.224
Only 3JCR (tri-snRNP)	0.581	0.629	-0.226
Only 5O9Z (B)	0.551	0.607	-0.255
ADD conformational-trajectory features [51]	n.s.	n.s.	+0.030, fails its degree-matched null ($p = 1.00$)
ADD protein-partner identity [52]	n.s.	n.s.	-0.026, does not transfer across genes

The gap between the nested single structure (0.597) and the oracle single structure (0.795) is the whole argument for the multi-state representation: a single state can be nearly as good as the ensemble, but only if you already know which gene you are testing, which in deployment you do not.

Two extensions were specified with kill criteria, tested and rejected. Both are reported because a negative result about an obvious extension is evidence about the problem, not an omission.

The first was a conformational-hinge hypothesis: that pathogenic variants sit at nucleotides whose protein partners turn over between consecutive states, rather than at the most buried ones. On the saturation screen, trajectory rewiring reached Spearman 0.150, but its partial correlation given burial was only +0.092 — below the +0.10 line declared in advance — while partner count given rewiring retained +0.299. Decisively, under a degree-matched permutation in which every position keeps its exact per-state partner count and only partner identity is shuffled, the null produced a *stronger* correlation than the real data (+0.281 $\pm$ 0.031 against +0.150, $p = 1.00$). The apparent +0.030 held-out gain resolved into a single gene with seven positives moving from 0.526 to 0.960 while two other genes got worse [51].

The second was partner identity, motivated by large per-protein Mantel-Haenszel odds ratios (PPIL2 31.8, SF3B1 22.2, PRPF8 7.7). Those are stratified by gene only, and pathogenic variants cluster *within* a gene, so any protein whose footprint covers the cluster earns a large ratio for free. Requiring the information to transfer to a held-out gene removes the confound, and it does not: held-out AUROC fell from 0.574 to 0.548, and identity alone reached 0.437, below chance. Only nine partners appear at labelled positions in three or more of the five labelled genes, and eight of those nine are the Sm ring, which binds every snRNA by construction [52]. We note that embedding partner proteins with a protein language model is the load-bearing component of comparable geometric-deep-learning methods in adjacent problems; it does not port here, and the reason is identifiable.

Four evaluation regimes

Table 3. Four evaluation regimes of increasing difficulty.

Regime	What	n	Positives	Performance
Held-out gene, dominant	Every variant of one gene withheld entirely	1,211	72	AUROC 0.766 [0.713, 0.811]
Held-out gene, recessive	Same, recessive positives	566	177	AUROC 0.591 [0.540, 0.639]
Held-out gene, pooled	Both mechanisms, rank-pooled	1,777	249	AUROC 0.636 [0.599, 0.672]
Zero-shot functional transfer	<i>RNU4-2</i> saturation genome editing; no functional measurement in training	485	continuous	Spearman 0.405 [0.255, 0.535]
Prospective, timestamped	Deposited 2026-10-09 with SHA-256, evaluation protocol fixed in advance	2,777	pending	to be evaluated on future reports

Held-out-gene is strictly harder than a cluster split: no variant, position or paralogue of the test gene is seen in training. In a temporal evaluation, trained only on evidence public before the ClinVar release of 4 May 2025 and only on other genes, the system ranked the 41 variants that subsequently became P/LP above CADD (pooled Δ AUROC +0.173, 0.039-0.304, $p = 0.009$; versus phyloP +0.111, $p = 0.11$) [17].

Population constraint and molecular function diverge

This is the study's principal finding, it began as a failed gate, and it is summarised in Figure 2.

Table 4. Zero-shot Spearman correlation with measured function (*RNU4-2* saturation genome editing, 485 variants).

Score	n	ρ	95% CI
Graph-smoothed contacts	485	0.437	[0.293, 0.561]
MuST-VEP	485	0.405	[0.255, 0.535]
Protein-contact fraction	485	0.347	[0.198, 0.482]
gnomAD allele density (± 2 nt)	485	0.258	[0.084, 0.410]
gnomAD constraint (1 - o/e)	485	0.206	[0.035, 0.344]
gnomAD allele density (position)	485	0.149	[-0.006, 0.287]
CADD (PHRED)	411	0.072	[-0.070, 0.208]
phyloP (447-way)	485	-0.004	[-0.129, 0.128]

Paired differences, all excluding zero: structure - constraint +0.231 (0.093-0.363, $p = 0.002$); structure - CADD +0.334 (0.144-0.525); structure - phyloP +0.440 (0.271-0.604). Partial correlations are asymmetric in both directions tested (Figure 2c): **graph-smoothed contacts retain ρ 0.394 controlling for constraint, whereas constraint retains 0.025 controlling for contacts**; for the combined MuST-VEP score the pair is 0.358 against 0.038. Among the 121 most functionally damaging variants by assay, the structural score places 59% in its own top quartile, against 45% for density, 40% for constraint, 21% for CADD and 11% for phyloP. Within the dominant critical region MuST-VEP reaches 0.575 (0.310-0.732).

The explanation is not subtle. Curated pathogenicity labels are assembled partly from population depletion, so a depletion score predicts them partly for free. A benchmark built on such labels will rank a population-derived score alongside a mechanistic one while the two carry different information about the molecule. This has been shown for missense variants, where gene identity dominates ClinVar benchmarks and function-based evaluation reorders predictors; we arrive second to that framing and say so. What we add is the non-coding case, where the effect is larger, and where — unusually — both label types exist for the same variants.

The practical consequence is narrow and worth stating precisely: for non-coding variants, computational evidence drawn from conservation or depletion is not independent of population-frequency evidence. Where a classification framework treats them as separate lines, it risks counting one observation twice.

Calibration against measured function

The 2026 guidance states that in silico tools should inform PP3/BP4 only once calibrated for snRNAs. Using the *RNU4-2* saturation screen as functional truth (damaging = lowest quartile of function score; 121 damaging, 364 tolerated across 137 positions) and zero-shot scores with the U4 family withheld, MuST-VEP reaches a positive likelihood ratio of 12.03 at its 90th percentile, with a position-block 95% CI lower bound of 4.87 — meeting the ClinGen Bayesian threshold for PP3_moderate on the conservative bound — at 33% sensitivity. The graph-smoothed contact feature alone reaches PP3_supporting at its 80th percentile (LR+ 7.41, lower bound 4.11, sensitivity 57%). CADD and phyloP447 reach no evidence band at any threshold tested (LR+ 0.50-1.04).

This is calibration against a cell-fitness readout, not against clinical outcome, and should not be substituted for clinical calibration. It is reported because it is the only functional truth set available for any snRNA gene, and because the alternative — calibrating on curated labels — is precisely the circularity documented above.

Allele-resolved scoring is a separate problem

The contact model scores positions, so all three substitutions at a nucleotide receive one score. We added allele-resolved features from the same structures [44] and tested them where the answer is measurable: within the 137 *RNU4-2* positions at which the screen assayed all three alternative alleles. A position-level score has, by construction, no within-position signal, so any correlation is attributable to allele identity. The allele score reached a mean within-position Spearman of +0.346 with -SGE (Wilcoxon $p = 0.0010$), and pair disruption alone +0.374; CADD, which is allele-resolved, reached -0.077. The most damaging allele was ranked first 44-46% of the time against 33% expected by chance, and 32% for CADD.

But adding those features to the clinical classifier made it worse: mean held-out AUROC for dominant variants fell from 0.810 to 0.785 (-0.026), driven by *RNU2-2* ($0.807 \rightarrow 0.716$), with no consistent gain for recessive variants ($+0.009$) [46]. Ranking alleles within a position and discriminating pathogenic from benign across positions are different tasks, and conflating them costs accuracy. We therefore report the allele score as a complementary output for the specific clinical question of which substitution at a known position is worst, not as a component of the classifier. The allele score varies only where the nucleotide is base-paired, which is 45 of 137 tested positions. This result rests on a single dataset and should be treated as promising and unreplicated.

Mechanism

Mantel-Haenszel odds ratios stratified by gene, at position level with BH-FDR [18a, 18b], placed dominant pathogenic variants at contacts with the activation and catalytic machinery — PRPF8 OR 6.4 (FDR 5×10^{-4}), RBM42 12.0 (0.003), SART1 5.8 (0.003), SNRNP200/Brr2 4.4 (0.006), SNRNP27 4.9 (0.009) — with the Sm ring not enriched (0.8). Recessive variants were enriched at snRNP-assembly contacts (SNU13 k-turn OR 4.4, FDR 0.036).

A pre-specified interaction test of this contrast did not meet its criterion: the joint Wald test over the three feature $\times$ mechanism terms gave $p = 0.022$, but the contact term ran opposite to the predicted direction and *RNU2-2* alone was flat ($p = 0.895$) [24]. We therefore report a gradient rather than a categorical separation. As shown in Table 2, these per-protein associations do not survive the requirement to transfer across genes, so they describe where pathogenic variants sit rather than providing independent predictive information.

Part of this gene family cannot be assessed at all

RNU1-3 and *RNU1-4* have zero callable alleles across 76,215 gnomAD genomes, while their near-identical siblings *RNU1-1* and *RNU1-2* are at full cohort depth. In the HPRC Release 2 pangenome, built from long-read haplotype assemblies, the same two genes carry 22 and 15 variants across 146 and 62 of 464 haplotypes — *RNU1-3* matching *RNU1-2* exactly at 134 variants per kb [41]. The cause is not poor coverage but non-identifiability: the four RNU1 copies are identical across all 164 nucleotides, so no short read can ever be assigned among them. The same holds for U6.

The same method distinguishes real absence from invisibility: of five genes returning zero variants of every class in a published systematic screen, three (*RNVU1-1* and two U7 genes) are genuinely near-invariant in long-read assemblies, and only *RNU1-3* and *RNU1-4* are not. *RNU4-2* itself shows 6 variants across 6 haplotypes, consistent with genuine constraint at a confirmed disease gene. Across coding sequence a comparable artefact exists but is small and already recognised: 89 of 34,556 MANE genes (0.26%) are missing at least half their expected synonymous variation, and only 2 carry any ClinVar P/LP variant [42]. Coding sequence has an internal control — synonymous variation — that non-coding sequence lacks.

U1 is the only major spliceosomal snRNA family with no established disease gene. We do not claim this is why. We claim only that the question has not been asked of those loci, cannot be asked with short reads, and can be asked with long reads.

Limitations

Two of four dominant genes contribute 7 and 4 pathogenic variants, so the dominant result rests substantially on *RNU4-2* and *RNU2-2*; the hierarchical intervals in Table 1 are correspondingly wide and several cross zero. The function validation is one assay in one gene, because *RNU4-2* is the only snRNA with a saturation screen. The allele result is unreplicated. The pangenome comparison uses 232 samples and graph-based genotypes that can err in repetitive sequence. GENCODE v27 models *RNU4-2* as 141 nt against the current 145, leaving four 3' nucleotides unscored. The model scores positions for indels, which matters because the single most common ReNU allele is an insertion. Recessive prediction does not work, for any method tested.

Conclusion

Structural context across the splicing cycle is, among the evidence types tested here, the only one that tracks what variants do to the molecule. Population-derived scores track what selection has done to the position, and on curated labels the two are hard to tell apart — which is why a pre-specified gate comparing them failed, and why that failure, once resolved against an ascertainment-free readout, became the study's main result. For a gene class where clinical interpretation is explicitly uncalibrated, that distinction decides what computational evidence is worth.

MuST-VEP provides scores for 13,280 variants across 17 spliceosomal snRNA genes, with within-gene percentiles, features, and an annotation of which loci population sequencing can and cannot assess. A set of 2,777 predictions was deposited with a cryptographic hash and a fixed evaluation protocol before any of the variants concerned were reported, and will be evaluated as new cases accumulate.

Figures

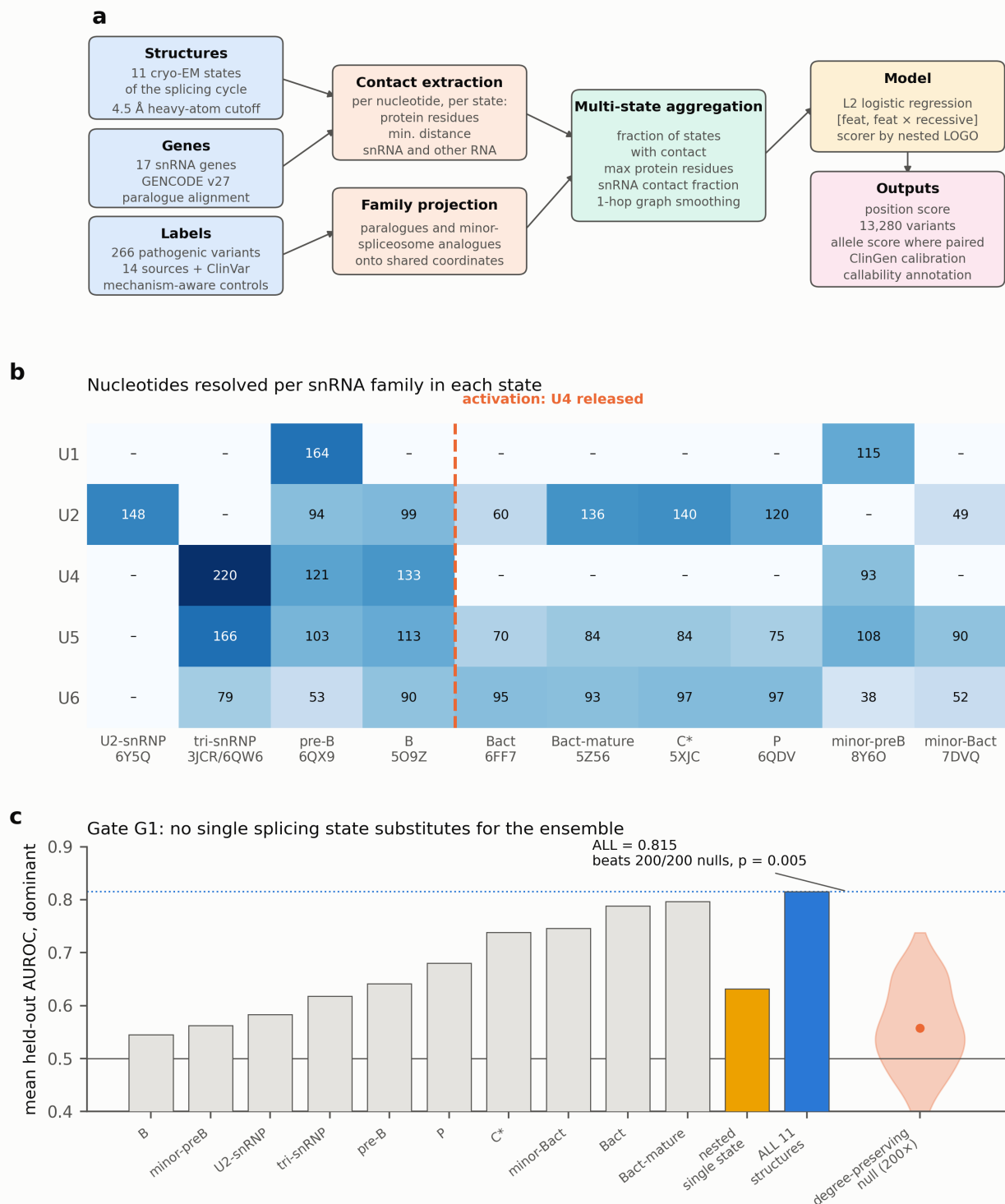


Figure 1. The MuST-VEP workflow and the case for a multi-state representation. (a) Pipeline: eleven cryo-EM states and 17 GENCODE snRNA genes feed contact extraction and family projection; multi-state aggregation with one-hop graph smoothing produces the features; a mechanism-aware L2 logistic model with nested leave-one-gene-out scorer selection produces position scores, allele scores where the nucleotide is paired, and a ClinGen Bayesian calibration. (b) Nucleotides resolved per snRNA family in each state. U4 is absent from every structure after activation, which is why no single state can serve *RNU4-2* and *RNU2-2* simultaneously. (c) Gate G1: mean held-out AUROC for dominant variants, each single state, the nested single-state selection, and the degree-preserving shuffled-contact null (200 replicates). The full ensemble reaches 0.815 and exceeds every replicate.

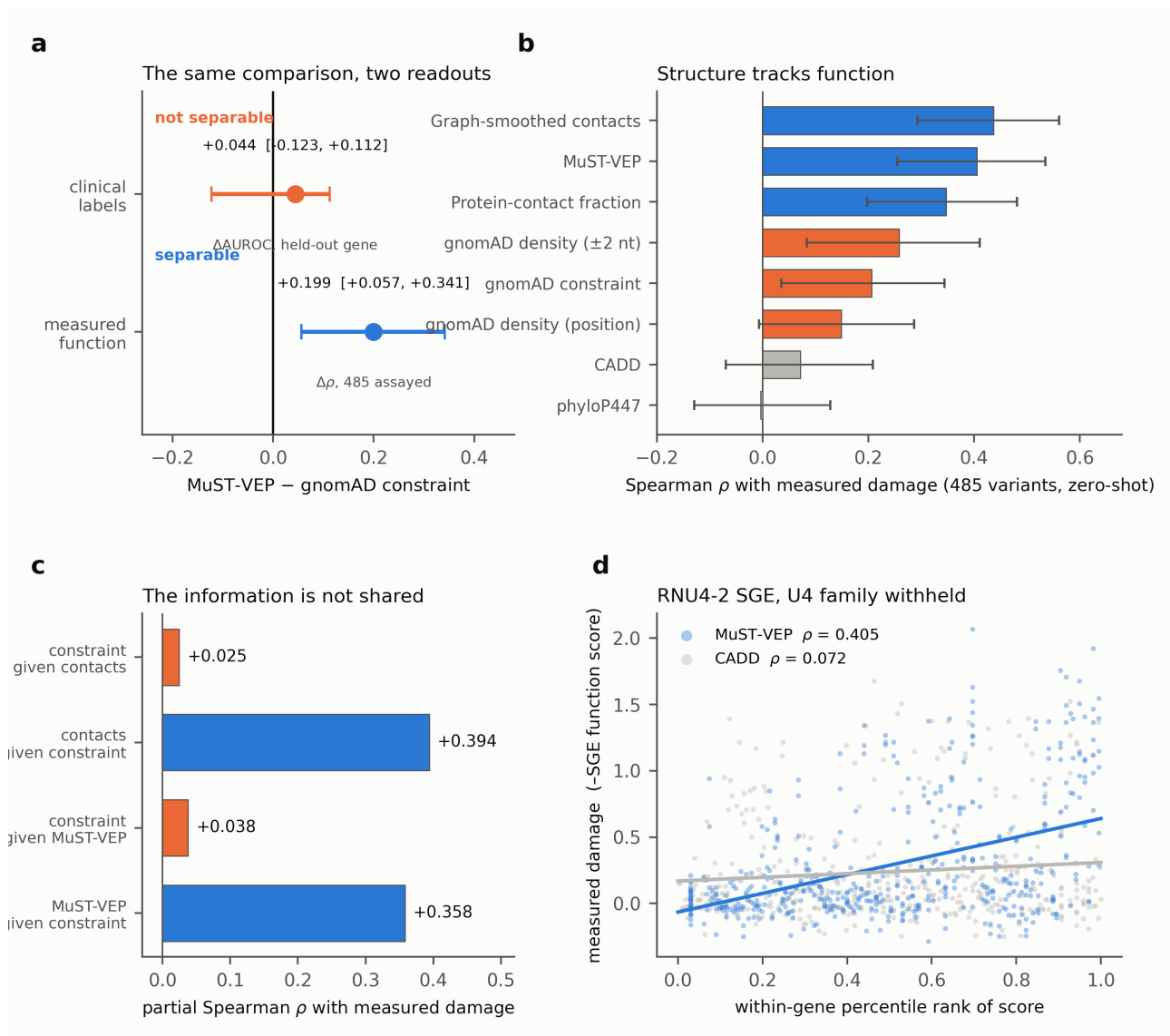


Figure 2. Population constraint and molecular function diverge. (a) The same comparison under two readouts, as paired differences rather than on a shared axis, because AUROC and Spearman are not commensurable: on clinical labels MuST-VEP – constraint is $+0.044$ [-0.123 , $+0.112$] and covers zero; on measured function it is $+0.199$ [$+0.057$, $+0.341$] and does not. (b) Zero-shot Spearman correlation with saturation-editing damage for every score, with position-block 95% intervals; structural features in blue, population-derived in orange. (c) Partial correlations, showing the information is not shared: graph-smoothed contacts retain 0.394 controlling for constraint, while constraint retains 0.025 controlling for contacts. (d) The underlying scatter for MuST-VEP and CADD on the same 485 variants, with the U4 family withheld from training.

Key points

existing in silico tools are not calibrated for non-coding RNA genes and notes that a single structure cannot capture how snRNA contacts change through the splicing cycle.

multi-state representation beats any single state chosen without access to the test gene (0.815 versus 0.631) and exceeds all 200 degree-preserving shuffled-contact nulls.

foundation model RNA-FM reaches 0.460, below chance.

correlations 0.025 versus 0.394 on saturation genome editing). Curated labels and molecular function are not interchangeable benchmarks for non-coding variants, because the labels are partly derived from population data.

kill criteria in advance, tested, and rejected; both are reported rather than omitted.

- Spliceosomal snRNA variants are a frequent cause of neurodevelopmental disorder, but current guidance states that
 - MuST-VEP represents each nucleotide by its contacts across eleven cryo-EM structures spanning the cycle. The
 - Against thirteen baselines under a held-out-gene protocol, MuST-VEP reaches 0.810 for dominant variants; the RNA
 - Population constraint matches the model on curated clinical labels but not on measured function (partial
 - Two obvious extensions — conformational-trajectory features and protein-partner identity — were specified with
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Author actions outstanding

1. Deposit the prospective prediction set (results/prospective_deposit_2026-10-09.tsv, SHA-256 b5b2249a...) on Zenodo and cite the DOI. 2. Generate the figures listed in FIGURE_PLAN.md, in particular **Figure 1**, the workflow schematic — both comparable papers in this venue lead with one and it does substantial work. 3. Send outreach email #1 in OUTREACH_DRAFTS.md to the saturation-editing group; time-sensitive. 4. Optional: AlphaGenome as a fourteenth baseline, which needs a free DeepMind API key.